

Meeting 3/03/2026

Summary

Action Items & Next Steps

- Students to develop a high-level design and architecture proposal for the multi-cloud performance evaluation system
- Students to create a POC to validate the proposed cluster deployment before proceeding to detailed design stage
- Students to develop a project timeline and plan for task distribution among team members
- Jenkins setup and pipeline configuration using VirtualBox — verify CI on merge with linting check
- Explore Docker fundamentals: networking, volumes, Scout, Compose — understand containers vs VMs
- LocalStack setup: run core AWS services (SQS, S3, DynamoDB, SNS) locally via boto3 and AWS CLI
- Kubernetes (Minikube) setup and exploration of pods, deployments, services, and namespaces
- Team to set up recurring weekly meetings and a shared communication channel

Student Progress Review

Work Completed (Week 1 - 3 March to 10 March):

- Jenkins setup and pipeline configuration using VirtualBox
- CI on merge implemented with linting check (static analysis hook on PR)
- Docker fundamentals studied: VMs vs containers, namespaces, cgroups, hypervisors
- AWS services overview via LocalStack (SQS, S3, DynamoDB, SNS, IAM)
- Kubernetes (Minikube) setup and exploration of basic K8s objects

Concepts Explored:

- Microservice code lifecycle and service boundary design
- Jenkins as the primary CI tool; Python/Ruby scripting for pipeline automation
- Build artifacts: lock files, metadata; uv-based Python package management
- Artifact repositories: JFrog (commercial) vs Sonatype Nexus (open-source); branch-level segmentation
- Git branching strategy: feature branch → PR → automated tests → merge
- SonarQube for code quality and security scanning
- Linux VM internals: storage, memory, network, compute managed by hypervisor
- Docker as lightweight VM alternative via Linux namespaces and cgroups

Architecture & Design Requirements

Reference Architecture Provided: HPE team shared guidance for architecture, not a final design. Students must develop their own high-level architecture for the multi-cloud cluster deployment.

Key Components in Reference:

- Microservices layer with FastAPI backend and React frontend
- CI/CD pipeline: Jenkins → SonarQube → Artifact repo → Container registry → K8s cluster
- LocalStack for AWS service emulation (dev) → real AWS (production)
- Observability stack: Grafana, Loki, Fluentd, Prometheus, Jaeger, OpenTelemetry
- Multi-cloud benchmarking layer: k6 / JMeter load testing across AWS, Azure, GCP

Design Considerations:

- Apply standard system design patterns — scalability and performance first
 - Map all identified technologies directly to the problem statement
 - Create component diagrams showing building blocks and their relationships
 - Consider cluster promotion chain as a core architectural constraint
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